

Software Requirements Specification (SRS)

Restaurant Management System (RMS)

Version: 1.0

Technology Stack

- Framework: CodeIgniter 4
- Database: PostgreSQL
- Frontend: Bootstrap 5
- Programming Language: PHP 8.x
- Architecture: MVC (Model-View-Controller)

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of the Restaurant Management System (RMS). The system is designed to automate restaurant operations such as menu management, table management, order processing, billing, and sales reporting.

This document serves as the primary reference for developers, testers, instructors, and stakeholders throughout the software development lifecycle.

1.2 Scope

The Restaurant Management System is a web-based application developed using CodeIgniter 4 and PostgreSQL. It aims to replace manual restaurant operations with a computerized system.

The system enables administrators to manage restaurant information, menu items, restaurant tables, customer orders, and billing while allowing cashiers to process customer orders efficiently.

The system includes:

- User authentication
- Dashboard
- Menu category management
- Menu item management
- Restaurant table management
- Customer order processing
- Billing and payment
- Daily and monthly sales reports

1.3 Intended Audience

This document is intended for:

- Software Developers
- Database Designers
- Testers
- Project Supervisors
- Restaurant Owners
- System Administrators

1.4 Definitions

Term	Description
RMS	Restaurant Management System
Admin	System administrator with full access
Cashier	User responsible for customer orders and billing
CRUD	Create, Read, Update, Delete

Term	Description
MVC	Model-View-Controller
PostgreSQL	Relational Database Management System

2. Overall Description

2.1 Product Perspective

The Restaurant Management System is an independent web application that automates daily restaurant operations.

The system stores all restaurant information in a PostgreSQL database and provides secure access through user authentication.

2.2 Product Functions

The system provides the following major functions:

- User Login
- User Logout
- Dashboard
- Menu Category Management
- Menu Item Management
- Restaurant Table Management
- Order Management
- Billing
- Receipt Generation
- Sales Reporting

2.3 User Classes

Administrator

Responsibilities

- Manage users
- Manage menu categories
- Manage menu items
- Manage restaurant tables
- View reports
- Configure the system

Cashier

Responsibilities

- Create customer orders
- Generate bills

- Receive payments
- Print receipts
- View today's sales

2.4 Operating Environment

Server Requirements

- Apache Web Server
- PHP 8.x
- CodeIgniter 4
- PostgreSQL 15+

Client Requirements

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

Operating Systems

- Windows
- Linux

2.5 Design Constraints

- Developed using CodeIgniter 4
- PostgreSQL database
- Bootstrap 5 interface
- MVC architecture
- Role-based authentication

3. System Features (Functional Requirements)

Module 1 User Authentication

Description

Allows authorized users to access the system securely.

Functional Requirements

FR-001 The system shall allow users to log in using a username and password.

FR-002 The system shall validate user credentials.

FR-003 The system shall deny invalid login attempts.

FR-004 The system shall allow users to log out securely.

FR-005 The system shall maintain user sessions until logout.

Module 2 Dashboard

Description

Displays an overview of restaurant activities.

Functional Requirements

FR-006 The system shall display the total number of menu items.

FR-007 The system shall display the total number of tables.

FR-008 The system shall display today's total orders.

FR-009 The system shall display today's total sales.

Module 3 Menu Category Management

Description

Allows administrators to manage food categories.

Functional Requirements

FR-010 The system shall allow administrators to add menu categories.

FR-011 The system shall allow administrators to edit menu categories.

FR-012 The system shall allow administrators to delete menu categories.

FR-013 The system shall display all menu categories.

Module 4 Menu Item Management

Description

Allows administrators to manage food items.

Functional Requirements

FR-014 The system shall allow administrators to add menu items.

FR-015 The system shall allow administrators to edit menu items.

FR-016 The system shall allow administrators to delete menu items.

FR-017 The system shall display all menu items.

FR-018 The system shall assign each menu item to a category.

FR-019 The system shall store the price of each menu item.

Module 5 Restaurant Table Management

Description

Allows administrators to manage restaurant tables.

Functional Requirements

FR-020 The system shall allow administrators to add restaurant tables.

FR-021 The system shall allow administrators to edit table information.

FR-022 The system shall delete unused tables.

FR-023 The system shall display table status as Available or Occupied.

Module 6 Order Management

Description

Allows cashiers to create customer orders.

Functional Requirements

FR-024 The system shall create new customer orders.

FR-025 The system shall assign orders to restaurant tables.

FR-026 The system shall add menu items to an order.

FR-027 The system shall update item quantities.

FR-028 The system shall remove menu items from an order.

FR-029 The system shall calculate the order total automatically.

FR-030 The system shall save completed orders.

Module 7 Billing

Description

Generates customer bills.

Functional Requirements

FR-031 The system shall generate customer invoices.

FR-032 The system shall calculate the total payable amount.

FR-033 The system shall record payment status.

FR-034 The system shall print receipts.

Module 8 Sales Reports

Description

Provides restaurant sales reports.

Functional Requirements

FR-035 The system shall generate daily sales reports.

FR-036 The system shall generate monthly sales reports.

FR-037 The system shall calculate total revenue.

FR-038 The system shall allow administrators to print reports.

4. External Interface Requirements

User Interface

The application shall provide:

- Responsive Bootstrap 5 interface
- Sidebar navigation
- Dashboard
- Data tables with search
- Pagination
- Confirmation dialogs
- Printable receipts

Hardware Interface

- Computer
- Thermal Receipt Printer (Optional)

Software Interface

- PHP 8.x
- Apache Web Server
- PostgreSQL
- CodeIgniter 4

5. Non-Functional Requirements

Performance

- Page loading time shall not exceed **3 seconds** under normal load.
- The system shall support at least **20 concurrent users**.

Security

- Passwords shall be securely hashed.
- Only authenticated users shall access the system.
- Role-based authorization shall restrict access to features based on user roles.
- User sessions shall expire automatically after a period of inactivity.

Reliability

- The system shall maintain data consistency during transactions.
- Failed operations shall not corrupt stored data.

Availability

- The system shall be available during restaurant operating hours.
- Planned maintenance shall be communicated to administrators.

Usability

- The interface shall be simple and intuitive for users with basic computer skills.
- Forms shall include validation and clear error messages.

Maintainability

- The application shall follow the MVC architecture of CodeIgniter 4.
- Source code shall be modular and documented.

6. Database Requirements

Main Tables

1. users
2. menu_categories
3. menu_items
4. restaurant_tables
5. orders
6. order_items

7. Business Rules

- Every menu item must belong to one menu category.
- Each restaurant table can have only one active order at a time.
- An order must contain at least one menu item.
- Bills are generated only for completed orders.
- Only administrators can manage menu categories, menu items, and tables.
- Cashiers can create orders, receive payments, and print receipts.
- Paid orders cannot be modified.

8. System Constraints

- The application must use CodeIgniter 4.
- PostgreSQL shall be the primary database.
- The system requires an internet or local network connection for multi-user access.
- The application shall run in modern web browsers.

9. Assumptions and Dependencies

- Users have valid login credentials.
- The web server and PostgreSQL database are installed and configured.

- The restaurant has at least one administrator account.
- Thermal printer support is optional.

10. Future Enhancements

The following features are outside the scope of Version 1.0 but may be included in future releases:

- Customer Management
- Table Reservation
- Inventory Management
- Supplier Management
- Kitchen Display System (KDS)
- QR Code Ordering
- Online Food Ordering
- Delivery Management
- Employee Management
- Loyalty and Rewards
- SMS and Email Notifications
- Multi-Branch Support
- REST API for Mobile Application